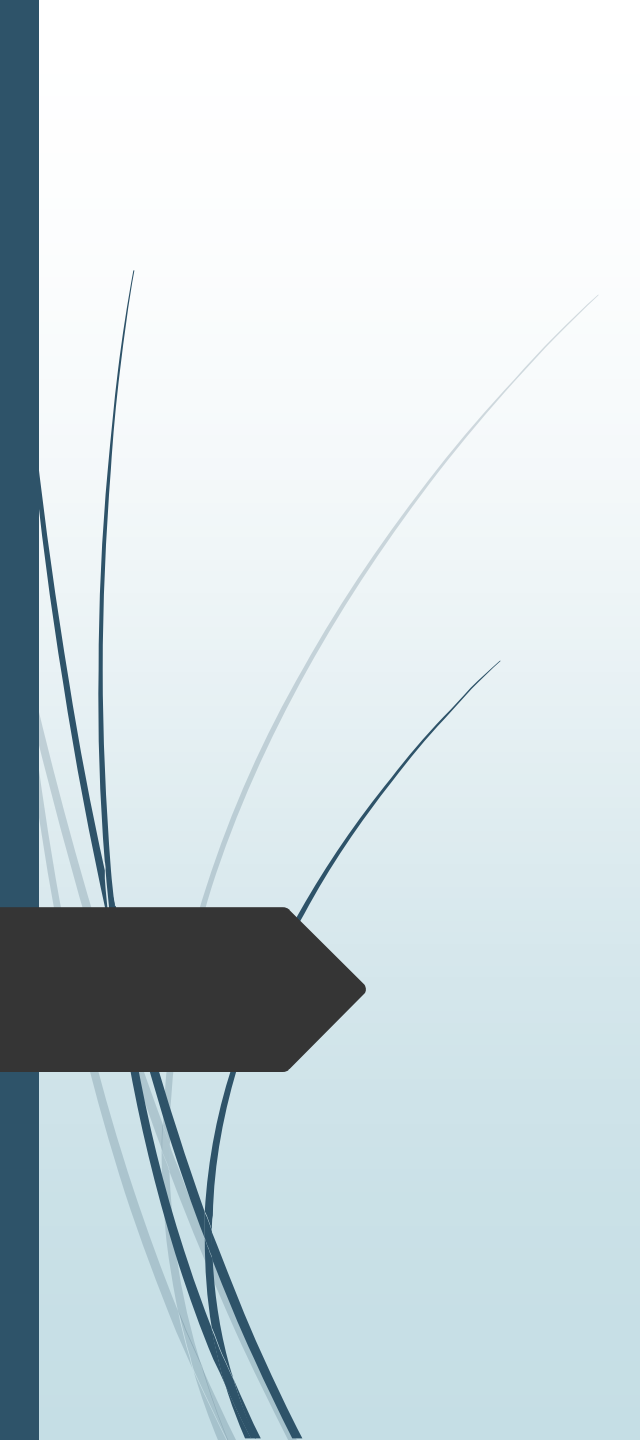
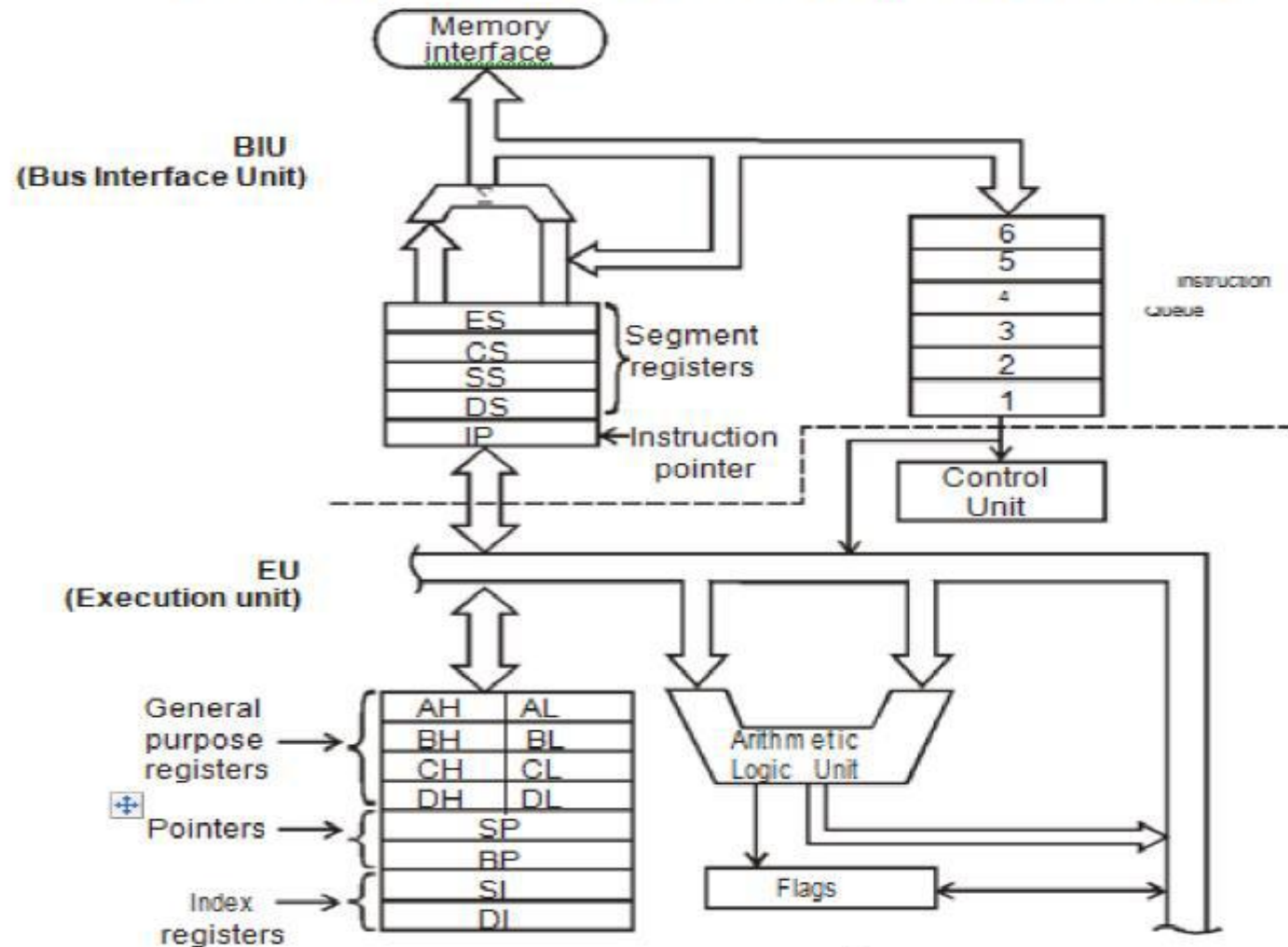




8086 Microprocessor Architecture

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8086 MICROPROCESSOR ARCHITECTURE





8086 Microprocessor Architecture

- The 8086 processor is partitioned logically into two processing units
 - **Bus Interface Unit (BIU)**
 - The BIU **fetches** instructions, reads data from memory and ports, and writes data to memory and I/O ports.
 - **Execution Unit(EU)**
 - EU receives program instruction codes and data from the BIU, **executes these instructions and stores the results** either in the general registers or output them through the BIU.



8086 Microprocessor Architecture

The BIU contains

- Segment registers
- Instruction pointer
- Instruction queue

The EU contains

- ALU
- General-purpose registers
- Index registers
- Pointers
- Flag register



Register

A register is **one of a small set of data holding places** that are part of a computer processor.

Registers are two types:

1. General-purpose or multipurpose Registers
 - These registers hold various data sizes and used for almost any purpose.
2. Special-purpose Registers
 - Used in special purposes.

Segment Register

- ❑ Segment Registers are **additional registers, generate memory addresses** when combined with other registers in the microprocessor.
- ❑ In a word, **segment registers are section of memory**

CS	CODE Segment
DS	DATA Segment
ES	EXTRA Segment
SS	STACK Segment



Segment Register

1. Code segment (CS)

- The code segment is a section of memory that **holds the code** (programs and procedures) used by the microprocessor.
- The code segment (CS) register defines the starting address of the section of memory holding **code**
- **The code segment size is limited to max 64KB.**



Segment Register

2. Data segment (DS)

- The data segment **contains most data** used by a program.
- The data segment (DS) register defines the starting address of the section of memory holding **data**
- The data segment size is limited to **max 64KB**.
- Data are accessed in the data segment by an offset address or the contents of other registers that hold the offset address.



Segment Register

3.Extra segment (ES)

- ES is an **additional data segment** used by some of the string instructions to hold destination data.
- The extra segment (register defines the starting address of the section of memory used as **extra segment**
- Can be used as an extra/additional segment for code or data.
- **Max size of the segment is limited to 64KB.**



Segment Register

4. Stack segment (SS)

- The stack segment defines the **area of memory used for the stack**
- The stack segment (SS) register defines the starting address of the section of memory used as stack
- The stack entry point is determined by the stack segment and stack pointer registers.

General-purpose or multipurpose Registers

- The MP has **07 general-purpose Registers**
- Each registers are **16 bit in size**, while some registers can be **divided into two 8-bit registers**
- The acceptable 8-bit register pairs are AH-AL, BH-BL, CH-CL, DH-DL. These pairs form AX, BX, CX, DX registers, respectively

AX	AH	AL	Accumulator
BX	BH	BL	Base index
CX	CH	CL	Count
DX	DH	DL	Data
	BP		
	SI		
	DI		

General-purpose or multipurpose Registers

1. AX register

- Called 16-bit accumulator, while AH/AL is 8 bit accumulator.
- Used for instructions such as **Multiplication division**.

2. BX register

- 16-bit Base index register, while BH/BL is 8 bits.
- Sometimes holds **offset address**.

3. CX register

- 16-bit Count register, while CH/CL is 8 bits.
- **Holds count for various instructions such as SHIFT, ROTATE, LOOP, etc.**

Offset is the displacement of the memory location from the starting location of the segment.



General-purpose or multipurpose Registers

4. DX register

- 16 bit Data register, while DH/DL is 8 bits.
- Holds a part of the result from multiplication or part of the dividend before a division.
- Need for I/O devices.

5. BP register

- 16 bit Base Pointer register.
- Points to a memory location.

6. SI register

- 16 bit Source Index register.
- Addresses source string data for the string instructions.



Special-purpose Registers

- Special-purpose registers include IP, SP, FLAGS.

1. IP register

- Called 16-bit Instruction Pointer.
- The Instruction pointer, **which points to the next instruction in a program is used** by the microprocessor to find the next sequential instruction in a program located within the code segment.

2. SP register

- 16-bit Stack Pointer, **addresses an area of memory called the stack**
- **The stack memory stores data through this pointer.**

FLAGS Register

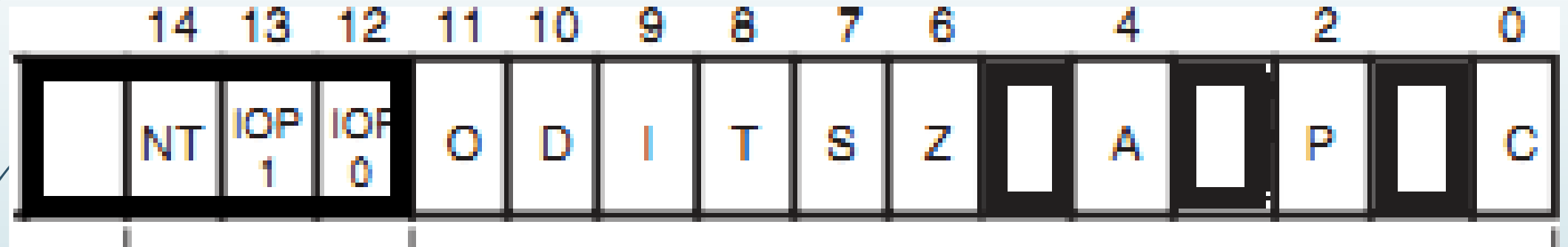
- ▶ A flag register (F or FL) is a 16-bit register, which indicates some condition produced by the execution of an instruction and controls certain operations of the MP
- ▶ **Flag register contains 09 active flags.** Among them 06 are status flags (indicate some condition produced by an instruction) and flags (control certain operations of processor). These are-

- ▶ Carry(C)
- ▶ Parity(P)
- ▶ Auxiliary carry(A)
- ▶ Zero(Z)

- ✓ Sign(S)
- ✓ Trap(T)
- ✓ Interrupt(I)
- ✓ Direction(D)

- ✓ Overflow(O)

FLAGS Registers...



FLAGS Registers...

1.C (Carry) Flag

- It holds the carry after addition or the borrow after subtraction.

$$\begin{array}{r} \text{FFH} \\ + \text{FFH} \\ \hline \text{FEH} \end{array}$$

$$\begin{array}{r} 1111\ 1111 \\ 1111\ 1111 \\ \hline 1\ 1111\ 1110 \end{array}$$

Carry

Here, **C=1**
If No carry, **C=0**

FLAGS Register

2.P (Parity) Flag

- Parity is the count of ones in register.
- Expressed as even or odd.
- Logic 0 for odd parity and logic 1 for even parity.
- If a number contains no one bits, it has even parity.

$$\begin{array}{r} \text{FFH} \\ + \text{FFH} \\ \hline \text{FEH} \end{array}$$

$$\begin{array}{r} 1111 \ 1111 \\ 1111 \ 1111 \\ \hline 1 \ 1111 \ 1110 \end{array}$$

Here, No. of " 1 " is 7
So, Odd parity i.e., P=0
If even carry, P=1

FLAGS Register

3.A (Auxiliary carry) Flag

► The auxiliary carry holds the carry (half carry) after addition or the borrow after subtraction between bits positions 3 and 4.

```
  FFH
+ FFH
-----
  FEH
```

4th position 3rd position

```
      1111 1111
      1111 1111
      -----
      1 1111 1110
```

The diagram illustrates the addition of two 8-bit values, FFH (1111 1111) and FFH (1111 1111). The result is FEH (1111 1110). The 4th position (bit 3) and 3rd position (bit 2) are highlighted. A carry of 1 is shown transferring from the 3rd position to the 4th position.

Here, “1” transfer between 3rd and 4th position bits. So, **A=1** otherwise, **A=0**



FLAGS register

5. S (Sign) Flag

- The sign flag **holds the arithmetic sign** after an arithmetic or a logical operation.
- If $S = 1$, the sign bit is set and the result is **negative (-ve)**.
- If $S = 0$, the sign bit is not set and the result is **positive (+ve)**.
- Usually sign bit is the leftmost bit.

6. T (Trap/ Trace) Flag

- It enables trapping through an on chip debugging (checking error) facility.
- $T = 1$, Debugging enable
- $T = 0$, Debugging disable

FLAGS register

7.I (Interrupt) Flag

- The interrupt flag controls the operations of the INTR (Interrupt request) input pin. If I = 1 the INTR pin is enabled; INTR pin is disabled with I = 0.

8.D (Direction) Flag

- The D flag selects either the increment or decrement mode for DI and/or SI registers during string instructions.
 - If D = 1, the registers are automatically decremented;
 - If D = 0, the registers are automatically incremented.
- Decrement mode indicates Set direction and write with STD (set direction) instruction Increment mode indicates Clear direction and write with CLD (clear direction) instruction



FLAGS register

9.O (Overflow) Flag

- An overflow indicates that the result has exceeded the capacity of the machine.
- Overflow occurs when signed numbers are added or subtracted.
- Mainly occurs during multiplication operation.
- If $O = 1$, Overflow occurs;
- If $O = 0$, No overflow.



Memory-addressing techniques

- Memory Address is the location of a program counter in a microprocessor.
- When a microprocessor is booted up, it loads the microcode that contains the addresses of the processor's registers.
- These registers are used to hold values that the microprocessor can read and write.
- The microprocessor places a value at the beginning of each register in the stack and places a value at the end of each register in the stack.
- When the microprocessor is called, it fetches the value at the location in the microcode that corresponds to the register on the stack.



Real Mode Memory Addressing

- Real mode operation allows MP to address only the first 1MB of memory space.
 - First 1MB of memory is called either real memory or conventional memory system.
 - DOS operating system requires MP to operate in real mode.
 - Windows does not use real mode.

Segment and Offset address

- A combination of a segment address and an offset address accesses a memory location in real mode. All real mode memory addresses must consist of segment address plus offset address.
- ✓ The segment address, located within one of the segment registers, defines the beginning address of any 64K byte memory segment.
- ✓ The offset address is also held in a register and selects any location within the 64K byte memory segment. An offset address is also sometimes referred to as displacement or logical address.
- A memory location pointed to by a segment address of 1000H and offset address of 2000H is written as 1000:2000.
- **What is the actual/physical memory location for 1000:2000**

Segment and Offset address

► Calculation of physical address from segment offset pair
1000:2000

1. Append **0H** to the right of segment value.
2. Add offset value with the appended segment value.
3. The result is the physical address for the given segment : offset pair

1	0	0	0	0
				0
				2
				0
				0
				0
				1
				2
				0
				0
				0

$$\text{Physical address} = \text{Segment add.} \times 10 + \text{Offset}$$

If Physical address is 22004H, what will be the Segment and Offset address?



Segment and Offset address

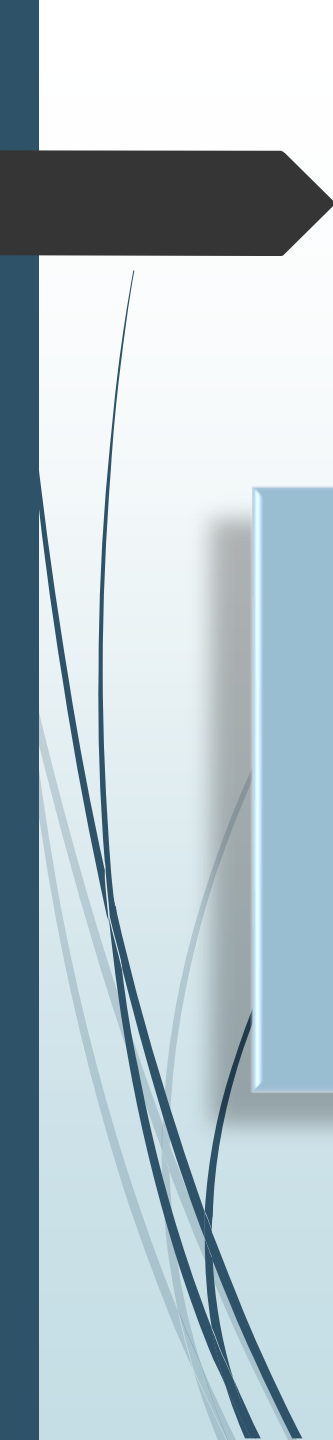
- Because a real mode segment of memory is 64K in length, once the beginning address is known, the ending address is found by adding FFFFH
- If segment register contains 3000 H, first address of segment is 30000H and last address is 30000 + FFFF or 3FFFFH

Default Segment and Offset Registers

- 8086 has a set of rules that apply to segments whenever memory is addressed. These rules define the segment and offset register combination.

- CS always used with IP to address next instruction in program CS register defines start of CS and IP locates next instruction within CS
- CS=1400H & IP=1200H MP fetches its next instruction from memory location 15200H.

Segment	Offset	Combination	Purpose
CS	IP	CS:IP	Code/ Instruction address
DS	BX,SI,DI	DS:BX/SI	Data address
ES	DI	ES:DI	String destination address
SS	SP or BP	SS:SP	Stack address



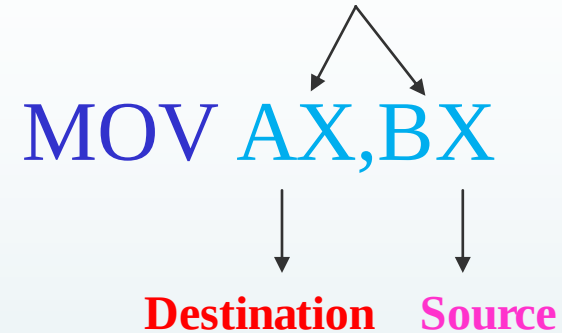
Addressing Modes of 8086



Addressing modes

- The different ways in which a processor can access data are referred to as addressing mode.
- Addressing modes for 8086 MP are three types
 1. Data addressing modes
 2. Program memory addressing modes
 3. Stack memory addressing modes

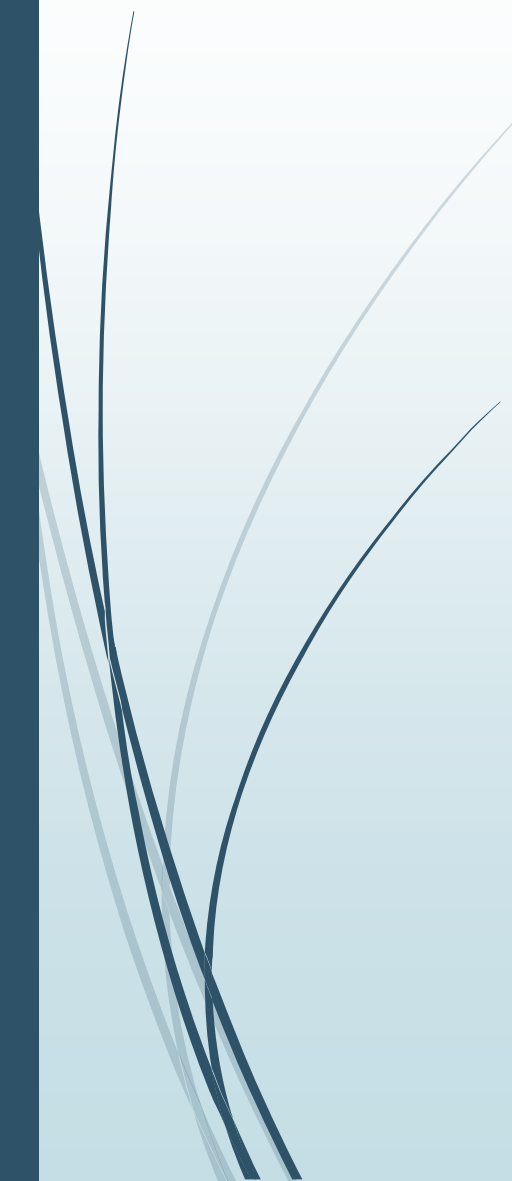
Addressing modes



- **MOV AX, BX** instruction translates the word contents of the **source register (BX)** into the **destination register (AX)**



Types of addressing mode in 8086

- 1. Immediate addressing mode
 - 2. Direct addressing mode
 - 3. Register addressing mode
 - 4. Register Indirect addressing mode
 - 5. Indexed addressing mode
 - 6. Register relative addressing mode
 - 7. Base plus index addressing mode
 - 8. Base relative plus index addressing mode
- 

1: Immediate addressing mode

In this type of mode, immediate data is part of instruction and appears in the form of successive byte or bytes

❖ **Not allowed**

- `MOV AH, 1000H`
- `MOV 75H, AL`
- `MOV CS, 1000H`

- Instruction is not allowed.
- Instruction is not allowed.
- Segment registers does not allow immediate data.

`MOV AX, 10ABH`

10	10AB _H	AX
----	-------------------	----